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Quantum Computing: Foundations, Architecture and Applications

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ABSTRACT

Quantum computing exploits the principles of quantum mechanics to address computational problems that are intractable to classical systems. This study examines the evolution, architecture, and applications of the field, with a focus on foundational principles, hardware advancements, and algorithmic progress. Recent quantum processors, such as Google's Willow and IBM's Heron, represent significant advancements in qubit count and gate fidelity; however, they remain constrained by qubit instability, environmental noise, and limitations of current error correction techniques. Quantum algorithms, including Shor's, Grover's, and HHL algorithms, have demonstrated substantial speedups in cryptography, optimization, and machine learning. Nevertheless, the realization of this potential in real-world problems encounters major bottlenecks related to low qubit counts and error correction. Applications span domains such as cryptography, drug discovery, precision medicine, financial modeling, and materials science, in which quantum computation offers potential breakthroughs. However, the development of practical quantum systems presents a substantial challenge. Key programming languages, such as Q#, Qiskit, and Cirq, facilitate algorithmic development and deployment; however, the efficiency of current quantum algorithms is limited by hardware constraints. The future of quantum computing lies in interdisciplinary collaboration, the development of resource-efficient error-correction techniques, and continued hardware development. This study underscores the potential of quantum computing, while emphasizing the research and development required to fully harness its capabilities to address major scientific and technological challenges.

1 | Introduction

Quantum computing employs foundational principles of quantum mechanics, including superposition and entanglement, to establish an advanced computational framework capable of performing complex calculations. Quantum bits (Qubits) represent both 0 and 1 concurrently, unlike conventional bits, hence allowing parallelism and exponential speed-up for certain tasks. From encryption to artificial intelligence to drug discovery to optimization, this technology has transforming power in many disciplines.

In this section, the historical perspective and the properties are quantum computing are discussed for the ease of better understanding of this article.

1.1 | Historical Perspective

The Babbage Engine showcased the principles of programmability and mechanical calculation. These principles laid the foundation to show the limitations of mechanical calculation and set forth the stage for a new computational paradigm that

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is quantum computing [1]. In the mid-20th century, electronic computers, such as ENIAC (1945), marked a significant milestone [2]. The years 1950s and 1960s saw smaller, faster, and more reliable transistor-based computers and the creation of high-level programming languages, such as FORTRAN and COBOL [3]. The 1970s introduced integrated circuits, reducing the size and cost of computers and leading to the rise of personal computers by companies such as Apple and IBM [4]. The 1980s and 1990s brought rapid advances in hardware and software, including graphical user interfaces (GUIs), the Internet, and powerful microprocessors [5], transforming computers into essential daily devices [6]. Despite these advances, classical computing struggles with complex problems that require exponential computational power. Quantum mechanics began in 1900 with Max Planck's concept of energy quantization [7]. This foundational idea revolutionized physics, setting the stage for subsequent discoveries such as Einstein's 1905 theory of special relativity and Bohr's 1913 model of the atom [8, 9]. The developments set the stage for quantum computers, a theory that deals with the smallest bits of matter.

Quantum mechanics' principles like superposition and entanglement enable non-classical computation. Qubits exist in superposition, enabling exponential scaling in some quantum algorithms. Entanglement allows for correlations between qubits to perform complex computations. These two properties are currently infeasible using classical computers. Early 20th-century quantum mechanics advancements laid the groundwork for quantum computing. Key developments included Planck's quantized energy levels in 1900, Einstein's particle-like light behavior in 1905 [8], Bohr's quantized electron orbits in 1913 [9], Schrödinger's wave mechanics and equation in 1926 [10], Heisenberg's Uncertainty Principle in 1927 [11], and Dirac's antimatter prediction in 1928 [12]. From the 1960s to the 1990s, quantum information theory emerged with milestones like Wiesner's quantum money concept in 1968 [13], and Brassard and Bennett's quantum cryptography protocol in 1984 [14]. By the mid-1980s, quantum models and algorithms were proposed, Feynman suggesting that only quantum computers could efficiently simulate quantum systems [15], and Deutsch proposing a general-purpose quantum computer in 1985 [16]. Notable algorithms, such as Shor's algorithm and Grover's technique, were introduced by Shor in 1994 and Grover in 1996, respectively [17], [18]. The first use of qubits in superconducting circuits occurred in 1999 with the development of superconducting qubits by Tsai, Yamamoto, and Nakamura [19], [20].

Classical computing continues to evolve with improvements in processing power, storage, and connectivity. Despite advances in Quantum Computing (QC), classical computers remain essential for many applications. Based on classical physics and Boolean logic, classical computers use bits (0 or 1) and process information sequentially. Although fundamental to modern technology, quantum computing offers a paradigm shift with potential in cryptography, optimization, and materials science.

1.2 | Quantum Computing

Quantum computing (QC) is a rapidly advancing field that seeks to leverage the principles of quantum mechanics to revolutionize computation. This technology is seen as a "grand scientific challenge" because of its potential to solve problems that

are intractable for classical computers. Recent advancements in quantum hardware and software have shown promising results toward achieving what is known as a "quantum advantage," where quantum computers perform computations faster than the best classical computers under practical conditions.

Recent breakthroughs in hardware, such as Google's 72-qubit quantum chip Sycamore, have significantly pushed the boundaries of quantum computing capabilities. Google's Sycamore demonstrated quantum supremacy by performing a specific quantum computation in 200 s, a task that would require the most powerful supercomputers approximately 10,000 years to complete [21]. Similarly, China's photonic quantum computer, Jiuzhang, has achieved quantum supremacy in Gaussian boson sampling, a benchmark significantly beyond the reach of current classical computers [22].

These advancements demonstrate how quantum phenomena like superposition and entanglement can be used by computers to solve challenging computational issues. The superposition allows qubits to be in multiple states at once, providing exponential growth in computational power with each added qubit. Entanglement enables qubits that are distant from each other to interact instantly, which is a critical feature for quantum algorithms.

In combination with artificial intelligence and machine learning, recent developments in quantum computing are increasingly interacting with medical imaging, therapeutic planning, and clinical decision-making. Quantum machine learning, for instance, has been used in medical decision-making where hybrid models combine QC and ML to uncover patterns from complicated datasets, optimize diagnosis and treatment processes, and allow individualized medicines [23]. Even under noise or restricted projection limitations, quantum-enhanced imaging techniques—such as tomographic reconstruction and quantum convolutional neural networks (QCNNs)—are starting to exhibit exceptional picture quality and stability [24]. Though still in their early years, these developing uses show the transforming power of QC combined with artificial intelligence in the medical field [25].

Despite these technological advances, significant challenges remain. Managing qubit coherence, minimizing errors caused by quantum noise and decoherence, and developing scalable fault-tolerant quantum systems are major hurdles [26]. Quantum systems are sensitive to external disturbances that can easily disrupt their quantum state, leading to errors. Therefore, enhancing the coherence time, that is, the time a qubit can maintain its quantum state, is crucial for the practical realization of quantum computing.

Furthermore, the development of effective quantum algorithms that can take advantage of the theoretical prowess of quantum machines is an ongoing area of intense research. Quantum algorithms like Shor's algorithm for factoring large numbers and Grover's algorithm for database searching promise substantial speed-ups over their classical counterparts, but implementing them on a practical scale involves complex engineering and theoretical challenges [27].

Numerous nations have initiated national quantum programs with the goal of expediting research and development in this domain, acknowledging the strategic significance of quantum technology. These initiatives aim to foster innovation, secure quantum communication networks, and establish a qualified workforce in quantum technologies [28].

Advancements in quantum computing have spurred a wave of investment from major players in the technology sector. Companies such as IBM, Google, and Microsoft are investing heavily in the development of quantum hardware and software, indicating a robust commercial interest in realizing the potential of quantum technologies.

2 | Quantum Information Processing

Quantum information is foundational for the operation of quantum computers, encapsulating the principles of encoding, processing, and manipulating quantum states. This field of science has witnessed rapid evolution because of significant advances in quantum computing, communication, and sensing, catalyzing developments in areas like prime number factoring and quantum chemistry. Qubits, the quantum equivalents of classical bits, are essential to understanding quantum information, since they can exist in numerous states concurrently, in contrast to their binary classical counterparts (being in state 0 or 1). This capability is due to quantum superposition and entanglement, two principal quantum mechanics phenomena [29].

Quantum superposition allows a qubit to be in a combination of states, amplifying the processing power of quantum computers by enabling them to perform multiple calculations simultaneously. Entanglement connects the states of two qubits even if they are separated by great distances, a feature that Einstein famously critiqued as “spooky action at a distance” [30]. These properties not only allow quantum computers to solve certain tasks much faster than classical systems but also offer new ways to process information that classical systems cannot emulate.

Classical computing relies on logical gates that operate under the laws of classical physics, manipulating bits in a linear, predictable fashion. In contrast, quantum gates manipulate qubits through operations that exploit their quantum mechanical properties, such as superposition and entanglement. This makes it possible for complex algorithms to be executed more quickly on quantum computers. For example, Shor’s algorithm for factoring large numbers can be executed exponentially faster on a quantum computer than it can be on the most advanced conventional computers [17].

Furthermore, while classical bits are constrained by the binary limits of their architecture, qubits operate under the broader, more complex rules of quantum mechanics. This distinction not only allows for the aforementioned speed advantages in computations such as integer factorization and unstructured searches, but also makes quantum computing potentially capable of tackling problems beyond the capabilities of classical machines, such as the simulation of extensive quantum systems [15].

Hence, the critical differences between classical and quantum information processing arise from their foundational principles:

Classical bits operate in a binary state governed by classical physics, whereas qubits exist in superpositions and are entangled, allowing parallel processing and operations that are guided by the laws of quantum mechanics, leading to potentially revolutionary computational speedups [31].

2.1 | Classical Bits versus Qubits

Classical bits and quantum bits, or qubits, serve as fundamental units of information in their respective computational paradigms but exhibit stark differences in representation and behavior. Classical bits are represented by binary digits, typically denoted 0s and 1s, existing exclusively in one of these two states at any given moment. An everyday analogy is a light switch that can be in either an “on” (1) or an “off” (0) position, representing a binary state.

In contrast, qubits operate under the principles of quantum mechanics, allowing them to exist in a superposition of states. The state of a qubit can be described by a linear combination of the basis states $|0\rangle$ and $|1\rangle$, such as $|0\rangle + |1\rangle$. This notation, where $|0\rangle$ and $|1\rangle$ are ket representations in the Dirac notation, indicates that the qubit is simultaneously in both states, a phenomenon not possible with classical bits. This superposition allows quantum computers to process information in parallel, vastly increasing their computational power over classical systems [32].

To better understand the practical implications, consider the task of searching for a specific item in an unsorted list. In classical computing, this would typically require a sequential search, checking each element one by one until the item is found, which means that the time to complete the search increases linearly with the size of the list. Quantum computing, however, can leverage Grover’s algorithm for this purpose. Grover’s algorithm uses the principle of superposition to search for all elements simultaneously and employs quantum interference to increase the likelihood of the correct result. This results in a quadratic speedup, making the search process faster than any classical algorithm could achieve [18].

The potential of quantum bits in computing is considerable, especially for tasks that benefit from parallel processing capabilities. Nevertheless, the practical implementation of qubits faces substantial hurdles. Among these challenges are decoherence and the correction of quantum errors. Decoherence occurs when qubits lose their quantum state because of interaction with the environment, effectively collapsing the superposition state back to a classical state. Error correction in quantum systems is also more complex than in classical systems because it must address errors in both the amplitude and phase of qubits [33].

2.2 | Quantum States

In quantum computing, information is encoded in quantum states associated with qubits. These states are typically represented using the ket notation, for example, $|\psi\rangle$, where ψ denotes the specific state of the qubit. Quantum states provide the most comprehensive description of a quantum system within quantum mechanics, represented as abstract mathematical entities

that fully describe a physical system. These states are expressed using Dirac or bra-ket notation as $|\psi\rangle$, with ψ being a variable that signifies the state [34].

In any quantum system, the quantum states can be defined in terms of basis vectors. For example, in a simple two-level qubit system, the basis vectors are $|0\rangle$ and $|1\rangle$. Extending this to an n -level system (or an n -dimensional Hilbert space), the set of basis vectors would include $|0\rangle, |1\rangle, \dots, |n-1\rangle$.

These basis vectors are subject to three crucial mathematical conditions:

1. **Normalization:** Each basis vector must have a length of 1. This condition is mathematically expressed as $\langle\psi|\psi\rangle = 1$. Normalization is crucial because it ensures that the probabilities of all possible outcomes sum up to 1, adhering to the probabilistic nature of quantum mechanics.
2. **Orthogonality:** The basis vectors must be orthogonal to one another, meaning the inner product of any two distinct vectors equals zero. Mathematically, this is denoted as $\langle\psi|\phi\rangle = 0$ for $\psi \neq \phi$. Orthogonality ensures that distinct quantum states are uncorrelated and independently observable.
3. **Completeness:** The set of basis vectors must span the entire space, implying that any arbitrary state $|\psi\rangle$ can be represented as a linear combination of these basis vectors. This is mathematically represented as: $|\psi\rangle = \sum_i c_i |i\rangle$, where c_i are complex coefficients, and the summation extends over all basis vectors in the system.

When these conditions are met, the basis is described as an orthonormal basis for the Hilbert space, which is an essential idea in quantum mechanics for characterizing the state spaces of quantum systems. These principles not only allow for a structured mathematical framework for quantum states but also facilitate the theoretical underpinnings necessary for quantum computation and various quantum algorithms. By understanding and manipulating these quantum states, quantum computers can perform operations that exploit phenomena like superposition and entanglement to achieve computational results beyond the reach of classical systems.

To better understand the fundamental principles governing quantum computing, it is crucial to delve into the mathematical representation of quantum states, with a particular emphasis on qubits, which are the fundamental units of quantum information. In quantum mechanics, the state of a qubit is often represented as a linear combination of basis states, denoted $|0\rangle$ and $|1\rangle$. These basis states form the backbone of quantum computation, encapsulating the possible states a qubit can assume.

2.2.1 | Mathematical Representation of Qubit States

A qubit can exist in one of the basis states, $|0\rangle$ or $|1\rangle$, or in a superposition of both. The concept of superposition is one of the key differentiators between classical and quantum computing. In a superpositional state, the qubit state is described by a weighted combination of the basis states, with each state's coefficient represented by a complex number. These coefficients,

when squared, provide the probabilities of the qubit collapsing to one of the basis states upon measurement. The sum of these probabilities must always be equal to one, satisfying the normalization condition essential for probability calculations in quantum mechanics [35].

2.2.2 | Examples of Qubit States

1. **Pure State $|0\rangle$:** When a qubit is solely in the $|0\rangle$ state, the coefficient for $|0\rangle$ is 1, and for $|1\rangle$ it is 0. Thus, the probability of finding the qubit in the $|0\rangle$ state is $|1|^2 = 1$, and in the $|1\rangle$ state is $|0|^2 = 0$, giving a total probability of 1.
2. **Pure State $|1\rangle$:** Similarly, if a qubit is solely in the $|1\rangle$ state, the probability calculation reverses: $|0|^2 = 0$ for the $|0\rangle$ state and $|1|^2 = 1$ for the $|1\rangle$ state, maintaining the total probability at 1.
3. **Equal Superposition State:** In an equal superposition, the qubit state can be represented as $(1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$. Here, each basis state's coefficient is $1/\sqrt{2}$, leading to a probability of $|(1/\sqrt{2})|^2 = 1/2$ for each state. The combined probability remains 1, verifying the normalization condition.

In more general terms, if the state of a qubit is expressed as $a|0\rangle + b|1\rangle$, where a and b are complex coefficients, the normalization condition requires $|a|^2 + |b|^2 = 1$. This condition ensures that the probabilities associated with all possible outcomes of measuring the qubit sum to 1, a fundamental axiom in quantum mechanics.

This precise mathematical framework allows quantum computers to exploit phenomena such as superposition and entanglement, offering computational capabilities far beyond those of classical systems. Such capabilities include performing parallel computations and executing algorithms like Shor and Grover, which are exponentially faster for specific tasks than their classical counterparts.

2.2.3 | Bloch Sphere

In quantum computing, the Bloch sphere serves as an essential geometric visualization tool to understand the complex state space of a single qubit. This three-dimensional model is incredibly useful for depicting the abstract concept of qubit states, which is fundamental to the principles of quantum mechanics and the technologies of QC [36].

The Bloch sphere is a unit sphere, where every point on its surface is indicative of a unique quantum state that the qubit can assume. Structurally, the north pole of the sphere is typically aligned with the basis state $|0\rangle$, and the south pole with the basis state $|1\rangle$. This orientation directly corresponds to the fundamental binary states of a qubit, analogous to the “0” and “1” in classical computing. The Bloch sphere's true value is revealed when it is employed to depict the behavior of quantum states under different operations. Quantum gates, essential for constructing quantum algorithms, can be visualized as rotations of the Bloch vector. This geometric perspective is crucial for grasping how various quantum gates influence a qubit's state—these transformations can be seen as straightforward rotations and reflections on the sphere.

2.3 | Quantum State Representation

Around the equator of the Bloch sphere, the represented states are those of superposition, where a qubit exists simultaneously in the states $|0\rangle$ and $|1\rangle$ to varying degrees, depending on the amplitude and phase of the specific quantum state. The relative positions along the equator—and indeed anywhere off the poles—indicate different superpositions, making the sphere a powerful tool for visualizing the continuum of quantum states.

2.3.1 | Analyzing Qubit Properties

The direction of the Bloch vector, starting from the center of the sphere and extending to the qubit's state on the surface, reveals information about the qubit's characteristics:

1. The angle between the Bloch vector and the z-axis gives an indication of the probabilities of the qubit being in the $|0\rangle$ or $|1\rangle$ state upon measurement.
2. The angle between the Bloch vector and the x-axis provides information on the relative phase between these basis states, which is crucial for understanding interference patterns and the behavior of quantum algorithms.

2.3.2 | Quantum Dynamics and Gate Operations

Quantum circuits are fundamentally constructed from quantum logic gates, which are essential for the processing of information within quantum systems. These gates function by modifying the quantum states of qubits, which serve as fundamental units of information in QC. Through various configurations of these gates, intricate computations are achievable, enabling the addressing of specific computational challenges. The development of quantum circuits and gates has progressed substantially, highlighting the importance of the quantum XOR or controlled NOT gate as a pivotal element [37]. Quantum logic gates, by incorporating quantum phenomena such as superposition and entanglement, enhance computational processes. These gates permit parallel processing and the simultaneous representation and manipulation of various states, which differ from classical gates due to the unique properties of quantum mechanics. They are quantum analogues of classical logic gates, but with some differences due to the quirks of quantum mechanics. Here are some types [38]:

- **Pauli Gates:** These include the X, Y, and Z gates. The X gate is a NOT gate (flipping $|0\rangle$ to $|1\rangle$ and vice versa), the Y gate applies a NOT gate and then changes the phase, and the Z gate changes the phase of $|1\rangle$. The applications of qubits to Pauli gates are illustrated in Figures 1 and 2.
- **Hadamard Gate (H):** This gate creates a superposition from a qubit in a definite state. It transforms $|0\rangle$ to $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$, and $|1\rangle$ to $\frac{|0\rangle-|1\rangle}{\sqrt{2}}$.
- **Phase Shift Gates:** These include S and T gates, which apply a phase shift of $\frac{\pi}{2}$ and $\frac{\pi}{4}$ respectively to the $|1\rangle$ state.
- **Identity Gate (I):** This gate leaves the state of the qubit unchanged.

- **CNOT Gate:** This is a two-qubit gate where the first qubit (the control qubit) flips the second qubit (the target qubit) if and only if the control qubit is $|1\rangle$.
- **Toffoli Gate:** This is a three-qubit gate where if the first two qubits (control qubits) are in state $|1\rangle$, the third qubit (target qubit) is flipped.
- **Swap Gate:** This is a two-qubit gate that swaps the state of the qubits, turning $|01\rangle$ to $|10\rangle$ and vice versa.

The Bloch-sphere representation of some of the basic quantum gates is illustrated here:

All these gates are unitary, meaning that they preserve the total probability of the system (the sum of probabilities of finding the quantum system in any of its possible states must always add up to 1) and are reversible, that is, can be undone by applying the inverse operation. They enable the alteration of qubits and facilitate the execution of algorithms on quantum computers. The field of QC is swiftly advancing and shows promising advancements in hardware and software. These advancements include the development of more stable qubits, improved error correction techniques, and the exploration of new quantum architectures. To overcome the challenge of decoherence, researchers are exploring various physical implementations of quantum logic gates, such as trapped ions, cavity QED, and N and P-based systems.

Furthermore, scientists are focusing on creating fault-tolerant quantum computing systems that can effectively mitigate the effects of noise and errors. In addition to hardware advancements, the development of efficient algorithms and software for quantum computing is crucial. These advancements will enable the realization of practical quantum computers that can solve complex problems with improved efficiency and accuracy.

2.4 | Quantum Measurement

Quantum measurement is a fundamental concept in quantum mechanics that sets it apart from classical physics. It refers to the process by which information about a quantum system is obtained through observation, and how this process influences the system's state. Unlike classical measurements, quantum measurements are inherently probabilistic and can alter the state of the system being measured [39].

A quantum system is described by a quantum state, represented mathematically by a wave function or state vector $|\psi\rangle$ in a complex Hilbert space. This state encapsulates all the information about the system and allows for the calculation of probabilities for various measurement outcomes. Physical quantities in quantum mechanics, such as position, momentum, energy, and spin, are represented by observable operators (Hermitian operators) acting on the Hilbert space. Each observable $\hat{A}$ has a set of eigenvalues $\{a_n\}$ and corresponding eigenstates $\{|a_n\rangle\}$ that satisfy the eigenvalue equation: $\hat{A}|a_n\rangle = a_n|a_n\rangle$.

The eigenvalues represent the possible results of measuring the observable, and the eigenstates are the states in which the system will be found after the measurement. Quantum mechanics introduces specific postulates related to measurement:

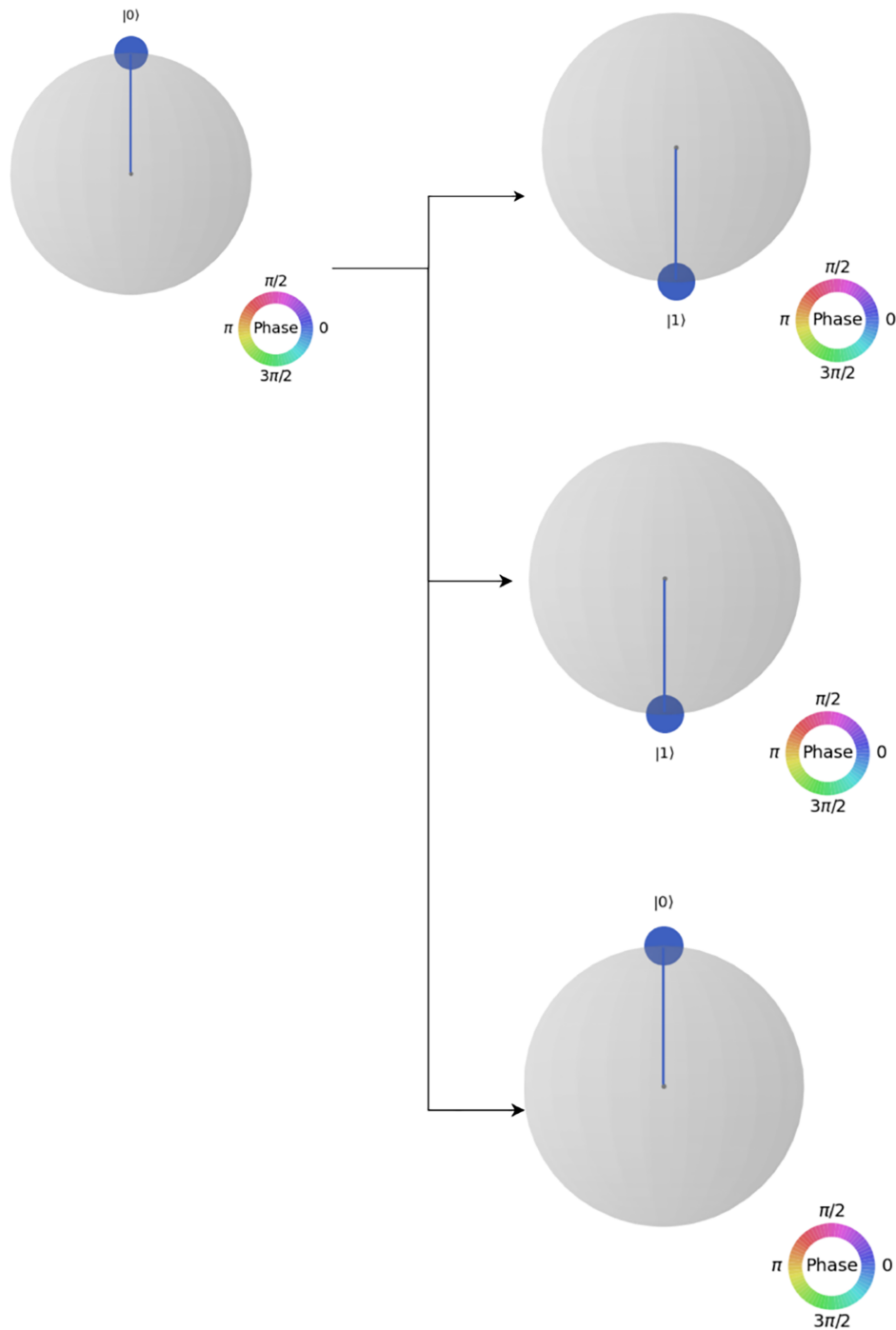


FIGURE 1 | Application of Pauli X, Y, and Z gates to a qubit $|0\rangle$. The Pauli X gate acts like a NOT gate for a qubit. When applied to a qubit in the $|0\rangle$ state (pointing up on the Bloch sphere), it flips the state to $|1\rangle$ (pointing down). This is depicted in the top diagram, where the vector flips from pointing up along the z-axis to pointing down. The Pauli Y gate not only flips $|0\rangle$ to $|1\rangle$ but also adds a phase factor of i . In the middle diagram, the vector rotates from up to down, and there's an accompanying phase shift annotation from 0 to $\pi/2$. This represents the added phase factor. The Pauli Z gate leaves $|0\rangle$ unchanged but would flip the phase of $|1\rangle$. This is shown in the bottom diagram where the vector remains pointing up, but there's an accompanying phase shift annotation from 0 to π , indicating the phase flip that would occur if the state were $|1\rangle$.

- Possible Outcomes: When measuring an observable $\hat{A}$, the only possible results are the eigenvalues a_n of $\hat{A}$.
- Probability of Outcomes: If the system is in state $|\psi\rangle$, the probability $P(a_n)$ of obtaining the eigenvalue a_n is given by the squared magnitude of the projection of $|\psi\rangle$ onto $|a_n\rangle$: $P(a_n) = |\langle a_n | \psi \rangle|^2$

- State Collapse: Immediately after the measurement yielding a_n , the system's state collapses to the corresponding eigenstate $|a_n\rangle$.

Prior to measurement, a quantum system can exist in a superposition of eigenstates. For instance, a state $|\psi\rangle$ may be expressed

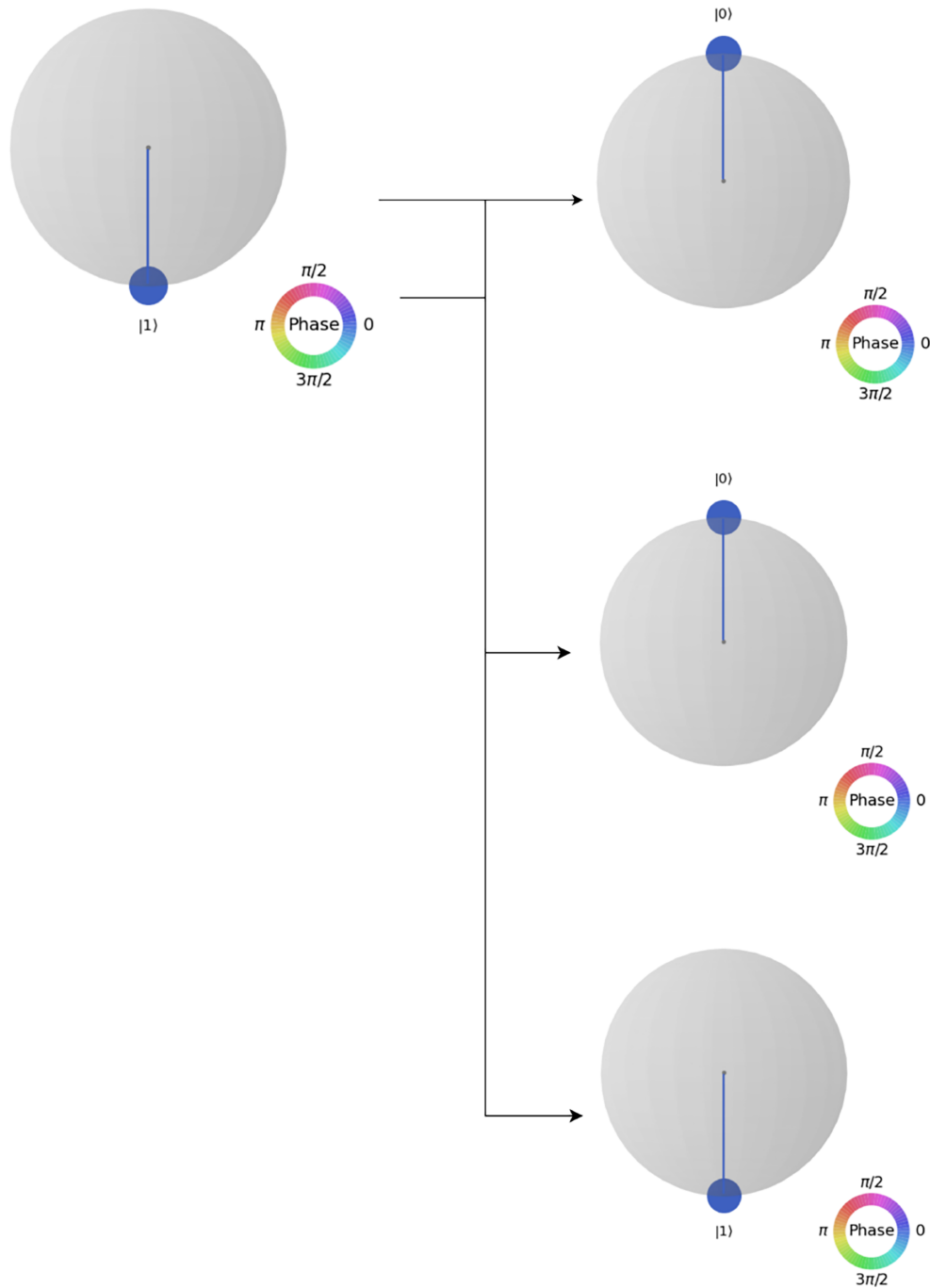


FIGURE 2 | Application of Pauli X, Y, and Z gates to a qubit $|1\rangle$. The Pauli X gate acts like a NOT gate for a qubit. When applied to a qubit in the $|1\rangle$ state (pointing down on the Bloch sphere), it flips the state to $|0\rangle$ (pointing up). This is depicted in the top diagram, where the vector flips from pointing down along the z-axis to pointing up. The Pauli Y gate not only flips $|1\rangle$ to $|0\rangle$ but also adds a phase factor of i . In the middle diagram, the vector rotates from down to up, and there's an accompanying phase shift annotation from π to $3\pi/2$. This represents the added phase factor. The Pauli Z gate leaves $|1\rangle$ unchanged but would flip the phase of $|0\rangle$. This is shown in the bottom diagram where the vector remains pointing down, but there's an accompanying phase shift annotation from π to $3\pi/2$, indicating the phase flip that would occur if the state were $|0\rangle$.

as a linear combination: $|\psi\rangle = \sum_n c_n |a_n\rangle$, where c_n are complex coefficients, and $\sum_n |c_n|^2 = 1$. During measurement, the act of measurement involves interaction between the quantum system and a measuring apparatus (often macroscopic and classical in nature). This interaction leads to the following outcomes: Entanglement, where the system and apparatus become entangled, meaning their states are correlated; and decoherence, where interaction with the environment causes the loss of quantum

coherence between different parts of the system's superposition, effectively selecting a particular outcome. After measurement: The system is found in one of the eigenstates $|a_n\rangle$ corresponding to the measured eigenvalue; the wave function has collapsed, and the superposition with respect to the measured observable no longer exists; and repeated measurements of the same observable will yield the same result, indicating the system remains in the eigenstate.

2.5 | Quantum Circuits

Quantum circuits are the foundational architecture for performing computations in the realm of quantum computing, much like classical circuits but with distinctly quantum features. In classical circuits, information is processed through a sequence of logical operations using bits that exist in one of two states: 0 or 1. Quantum circuits, on the other hand, operate on quantum bits, or qubits, that can exist in a superposition of states [40].

A qubit, the basic unit of quantum information, is represented by a two-level quantum system, typically denoted as $|0\rangle$ and $|1\rangle$, which correspond to the classical bit states, but a qubit can also be in any linear combination, known as a superposition, of these states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

where α and β are complex numbers such that $|\alpha|^2 + |\beta|^2 = 1$, due to the normalization condition of the quantum states.

Quantum circuits are made up of quantum gates, which are the building blocks that manipulate qubits. Each gate is represented by a unitary operator, ensuring that the quantum evolution is reversible and preserves the total probability. A simple quantum gate is the Hadamard gate (H), which operates on a single qubit and creates a superposition state:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (2)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (3)$$

Another fundamental gate is the Pauli-X gate, which is analogous to the classical NOT operation, flipping the state of a qubit:

$$X|0\rangle = |1\rangle \quad (4)$$

$$X|1\rangle = |0\rangle \quad (5)$$

Quantum circuits can also have multi-qubit gates, such as the Controlled NOT gate, which performs a NOT operation on the target qubit only when the control qubit is in the $|1\rangle$ state:

$$CNOT|00\rangle = |00\rangle \quad (6)$$

$$CNOT|01\rangle = |01\rangle \quad (7)$$

$$CNOT|10\rangle = |11\rangle \quad (8)$$

$$CNOT|11\rangle = |10\rangle \quad (9)$$

These operations exhibit the fundamental property of quantum mechanics, the superposition principle, by mapping the basis states into superpositions. Moreover, quantum gates can also yield entanglement, another key feature of quantum mechanics, which enables a set of qubits to have states that are inseparably correlated.

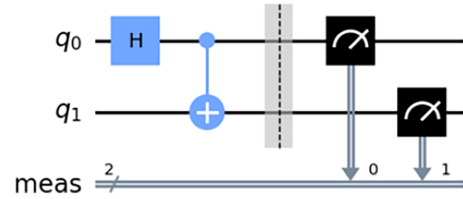


FIGURE 3 | The quantum circuit illustrates a basic two-qubit system with the Hadamard and CNOT gates. The Hadamard gate (denoted as “H”) is applied to the first qubit (q_0). This gate creates a superposition state, where q_0 is in an equal mixture of $|0\rangle$ and $|1\rangle$ states. After the Hadamard gate, there is a CNOT gate with q_0 as the control qubit and q_1 as the target qubit. The CNOT gate entangles the two qubits, meaning the state of q_1 will flip from $|0\rangle$ to $|1\rangle$ (or vice versa) only if q_0 is in the $|1\rangle$ state. Both qubits are then measured, as indicated by the meter symbols on q_0 and q_1 . The outcome of these measurements collapses the quantum state into classical bits.

To illustrate how gates are used in sequence, consider a two-qubit quantum circuit using a Hadamard gate followed by a CNOT gate as illustrated in Figure 3. First, we apply the Hadamard gate to the first qubit, which is initially in the $|0\rangle$ state:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (10)$$

After applying the Hadamard gate, the system is in a superposition of the $|0\rangle$ and $|1\rangle$ states for the first qubit. Next, we apply the CNOT gate with the first qubit as a control and the second qubit (also initially in the $|0\rangle$ state) as the target:

$$CNOT\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle\right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (11)$$

After the application of the CNOT gate, the two qubits are now entangled. The resulting state cannot be separated into individual qubit states, which means that the condition of one qubit immediately influences the condition of the other, no matter how far apart they are, showcasing the phenomenon of entanglement.

We plot the histogram of the measurement results to illustrate the entanglement phenomenon, as shown in Figure 4. The output state vector after applying the Hadamard gate followed by the CNOT gate should be close to $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, demonstrating the entanglement between the two qubits. Quantum circuits can perform many algorithms that are more efficient than their classical counterparts. A prominent algorithm is the Quantum Fourier Transform (QFT), which converts a quantum state into its frequency representation and is essential for various quantum algorithms, such as Shor’s algorithm for prime factorization. The QFT is applied to a register of qubits and is defined by the transformation:

$$QFT|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k\rangle \quad (12)$$

where $N = 2^n$ is the dimension of the state space of the qubit register, and j and k are integers representing the quantum states in binary form.

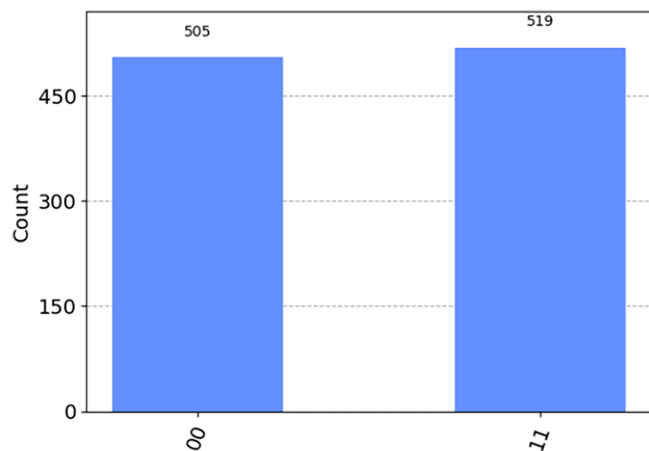


FIGURE 4 | The histogram provided shows two peaks, one for the state $|00\rangle$ and another for the state $|11\rangle$, with counts of approximately 505 and 519, respectively. This distribution of measurement results aligns with the expected outcome of the quantum circuit after applying a Hadamard gate followed by a CNOT gate.

The design of quantum circuits requires careful assembly of quantum gates, ensuring that the final quantum state after the operation of the circuit performs the desired computation. Crucial to all this is error correction, as quantum states are fragile and prone to decoherence. Quantum error-correcting codes, such as the Shor code, use additional auxiliary qubits to diagnose and correct errors without collapsing the computation.

Understanding quantum computing gates and circuits is key to the field. Gates like Pauli-X, Hadamard, and CNOT are the building blocks for manipulating qubits in complex computations. These operate under superposition and entanglement, unique to quantum mechanics, offering exponential speedup for certain problems over classical computing.

2.6 | Hardware Components of Quantum Computers

Quantum computing systems necessitate meticulously synchronized components to facilitate interactions with users, data, and networks, often relying on classical computing resources to enhance operational efficiency. These systems are architecturally structured into four fundamental hardware layers. The quantum data plane serves as the foundational layer, housing the qubits and enabling their operational capabilities [41]. The control and measurement plane is tasked with the execution and precise measurement of qubit operations as dictated by computational requirements. The control processor plane orchestrates the sequence of operations and measurements in alignment with the algorithmic framework, utilizing measurement outcomes to guide subsequent quantum tasks. Finally, the host processor, which operates as a classical computer, governs user interaction, network communication, and data storage through a conventional operating system and user interface. This host processor is interconnected with the control processor via a high-bandwidth link to ensure seamless system functionality [42].

2.6.1 | Quantum Data Plane

The quantum data plane constitutes the core of a quantum computer, housing physical qubits and the infrastructure essential for their operation and maintenance. It includes circuitry for measuring qubit states, executing gate operations in gate-based systems, and managing the Hamiltonian in analog frameworks. In gate-based architectures, this layer enables qubit interactions via programmable wiring, while analog systems require more intricate qubit couplings. Achieving high qubit fidelity necessitates robust environmental isolation, which restricts interqubit connectivity and imposes computational constraints within the system's architecture. Key performance metrics include gate error rates, interqubit connectivity, coherence times, and qubit density per module. Unlike classical systems, quantum data planes rely on external control mechanisms, employing analog signals delivered through wires, optical, or microwave emissions, requiring precise management to mitigate interference. As qubit counts increase, ensuring accurate signal transmission grows more challenging, making module capacity a critical design consideration [43].

2.6.2 | Control and Measurement Plane

The control and measurement plane bridges the quantum data plane and the control processor, translating digital signals from the processor into analog signals for quantum operations and converting qubit measurements into classical binary data. Accurate signal generation is challenging due to the analog nature of quantum gates, where minor inaccuracies or qubit design flaws can accumulate, reducing operational precision. Signal crosstalk, caused by insufficient isolation, can also disrupt qubit states. These issues are mitigated through continuous calibration, refined pulse shaping, and software adjustments. As qubit counts increase, calibration complexity escalates due to signal interactions. Different qubit technologies require distinct control signals: Trapped-ion qubits utilize microwave or optical signals, while superconducting qubits rely on microwave and low-frequency electrical signals delivered via cryogenic systems. Quantum operations demand precise signal delivery, facilitated by advanced classical signal generators. Gate speeds vary significantly, ranging from tens to hundreds of nanoseconds for superconducting qubits to up to 100 μ s for trapped-ion qubits. While silicon technology provides sufficient precision, the quantum data plane remains the primary bottleneck for gate speed [43].

2.6.3 | Control Processor Plane

The control processor plane orchestrates quantum algorithm execution by determining the appropriate Hamiltonian or sequence of quantum gate operations and measurements. Instructions from the host processor are adapted to the quantum layer's capabilities via a software tool stack. A critical function of this plane is implementing quantum error correction algorithms, which demand substantial classical computation to address errors based on syndrome data. Timely error correction is essential, as delays can disrupt quantum operations. To manage computational demands, which grow with qubit count and error-correcting code complexity, this plane may employ multiple networked

processors. Scalable designs for large quantum systems often divide the control processor plane into a classical processor executing the quantum program and dedicated hardware interfacing with the control and measurement plane to process syndrome data and compute subsequent qubit operations. Key challenges include developing scalable, high-speed hardware and creating effective high-level abstractions. Users interact indirectly through the host processor, which acts as an accelerator, translating compiled code into commands for the control and measurement layer. Similar to classical accelerators for graphics or machine learning, it connects to the host processor via high-bandwidth memory for efficient data and program transfer [44].

2.6.4 | Host Processor

The host processor functions as a classical computer equipped with a standard operating system and libraries, offering well-known tools and services for software development. It facilitates the creation of applications specifically designed for quantum processors, which are distinct from those used in classical computing, and provides the necessary storage and networking features crucial for quantum applications. By combining a quantum processor with a classical computer, the system takes advantage of current classical computing capabilities, eliminating the necessity to construct completely new infrastructure.

2.7 | Quantum Computers

Quantum computers tackle certain problems much quicker than conventional computers by using qubits to execute computations grounded in the ideas of quantum physics. For jobs requiring complicated optimization, large-scale simulations, and so on, they are very potent. In this section, the quantum computer architecture, the qubit technologies, and the real-time quantum processors available in recent years are discussed in detail.

2.7.1 | Quantum Computer Architecture

An effective quantum computer architecture must combine simplicity with the capability to manage the complex resources inherent in quantum computing. A layered approach is a standard solution in engineering, including information technology, to address such challenges [45]. Quantum computer architecture typically comprises five key layers: The Physical Layer, Virtual Layer, Quantum Error Correction Layer, Logical Layer, and Application Layer. Figure 5 shows the Five-layered Quantum Computer Architecture [46].

The Physical Layer serves as the foundation of the architecture, containing physical qubits, the associated hardware, and the environment that maintains their quantum states. Quantum gates are implemented at this layer through various physical operations. The Physical Layer interfaces with the Virtual Layer to manage qubit readouts and gate operations.

The Virtual Layer transforms quantum states into informational primitives such as qubits and quantum registers. This layer

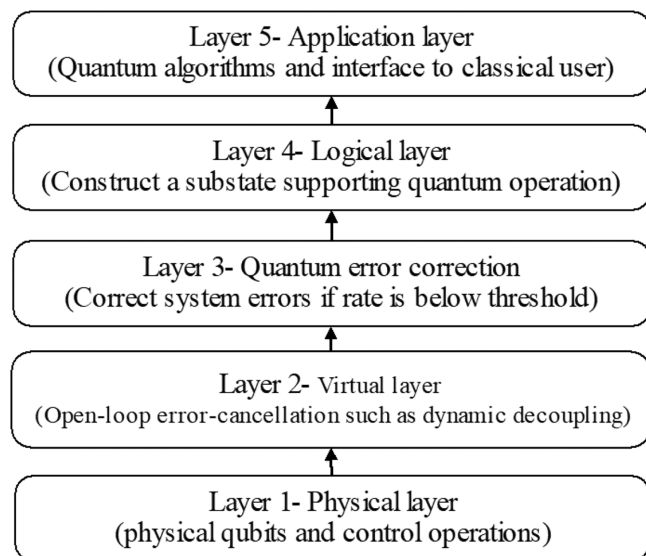


FIGURE 5 | Five-layered Quantum Computer Architecture.

converts physical operations into logical gates, helping to establish a coherent quantum system. It provides virtual qubits, logic gates, and measurements for integration with the Quantum Error Correction Layer.

The Quantum Error Correction Layer is critical for mitigating errors caused by noise, ensuring fault-tolerance in high-capacity quantum computers. Using Quantum Error Correction (QEC) technology, this layer identifies and corrects errors by extracting entropy as error syndromes. It produces logical qubits and logical gates, which are then passed to the Logical Layer.

The Logical Layer abstracts fault-tolerant structures of logical qubits and gates, enabling the creation of arbitrary gates for use in the Application Layer. This abstraction is crucial for realizing a universal quantum computer.

The Application Layer is the user-facing interface where quantum algorithms are executed. Users interact with the system using quantum-encoded input data. Independent of the underlying layers, the Application Layer operates on application qubits and arbitrary logic gates, making it highly flexible for diverse quantum computing applications. This layered architecture efficiently organizes the intricate resources of quantum computers, paving the way for scalable and reliable quantum computing systems.

2.7.2 | Qubit Technologies

Quantum computers are being developed through various approaches, with five main qubit technologies leading the field [21]:

1. Superconducting Qubits: Superconducting quantum computers are among the most widely adopted. These systems use superconducting materials and tiny electrical circuits to produce and manipulate qubits, enabling fast gate

operations. Key players in this space include Google, IBM, IQM, and Rigetti Computing [47].

2. **Photonic Qubits:** Photonic quantum computers leverage photons (light particles) to process quantum information. This approach, which avoids cryogenic or laser cooling required by other systems, is a promising alternative for large-scale quantum computing. Companies like Xanadu, ORCA Computing, Quantum Computing Inc., and PsiQuantum are prominent in this area, each exploring unique techniques such as Xanadu's "squeezed light" and PsiQuantum's distinct methods [48].
3. **Neutral Atoms:** Neutral atom quantum computers trap atoms in an ultrahigh vacuum using tightly focused laser beams, known as optical tweezers. These systems are less sensitive to stray electric fields, offering robust options for quantum processors. Leading companies include Pasqal, Atom Computing, ColdQuanta, and QuEra [49].
4. **Trapped Ions:** Trapped ion quantum computers use charged atoms or molecules (ions) confined with electric and magnetic fields to store and process quantum information. These systems boast stability and long coherence times, making them suitable for precise applications. Notable companies in this domain include Quantinuum, IonQ, Alpine Quantum Technologies, and eleQtron [50].
5. **Quantum Dots:** Quantum dot quantum computers rely on silicon-based qubits formed from pairs of quantum dots. This technology holds promise for developing robust and scalable qubits. Companies like Diraq, Siquance, and Quantum Motion are actively working in this field [51].

2.7.3 | Quantum Processors

Quantum processors, also referred to as quantum processing units (QPUs), are challenging to compare because of the diverse architectures and methodologies employed in their design. The published number of physical qubits alone does not accurately represent the performance capabilities of a processor, as various factors influence their efficiency and reliability. Some of the listed quantum processors have seen significant attention but lack comprehensive demonstrations or scientific publications detailing their performance characteristics.

1. **IBM Quantum Processors:** IBM continues to lead quantum computing innovation with advanced heavy-hexagonal qubit architectures and cutting-edge error correction techniques. The IBM Hummingbird processor, released in 2020, features 65 qubits, achieving a quantum volume of 128 and introducing scalability enhancements like flip-chip technology and readout multiplexing, with improved coherence in subsequent revisions. IBM Eagle, unveiled in 2021, marked IBM's first processor to surpass 100 qubits, offering 127 qubits with error correction innovations like Hamiltonian tomography, demonstrating potential for large-scale quantum applications. IBM Heron, released in 2023, introduced 133 qubits with a quantum volume of 512 and further advancements in tunable coupler architecture for error correction, later expanded to 156 qubits with TLS mitigation in 2024. IBM Condor, the largest processor to

date with 1121 qubits, highlights IBM's ambition to push scalability, though its performance remains under research. These advancements underscore IBM's dedication to building powerful quantum processors for increasingly complex computations [52].

2. **Intel Quantum Processors:** Intel has been actively advancing quantum hardware with a focus on scalable, reliable architectures, showcasing innovation in both superconducting and silicon spin qubit technologies. Tangle Lake, unveiled at the Consumer Electronics Show in 2018, marked Intel's significant entry into quantum computing with 49 superconducting qubits [53]. This processor operates at extremely cold temperatures of about 20 millikelvin and employs the "surface code" for error correction, leveraging loops of superconducting metal in its architecture. Tunnel Falls, announced on June 15, 2023, represents Intel's latest advancement, featuring 12 silicon spin qubits that are up to 1 million times smaller than other qubit types, enhancing scalability [54]. Fabricated on 300-millimeter wafers for high yield and uniformity, Tunnel Falls utilizes Intel's expertise in transistor fabrication and is in the early stages of quantum error-correction development to ensure reliability as qubit counts increase. These advancements underscore Intel's commitment to leveraging its semiconductor expertise for scalable and practical quantum computing systems.
3. **Google Quantum Processors:** Google has pioneered advancements in quantum computing with its focus on superconducting qubit architectures and innovative error correction methods, such as the surface code method AlphaQubit. In 2018, Google introduced Bristlecone, a 72-qubit processor with a square grid architecture designed to test error rates and scalability using a 9-qubit linear array for error correction. The following year, Sycamore, a 53-qubit processor, achieved quantum supremacy by completing a task in seconds that would take classical supercomputers millennia, while also initiating foundational experiments in error correction. In 2024, Google unveiled Willow, a 105-qubit processor with enhanced qubit coherence and gate fidelity, demonstrating exponential error correction with rates 20 times better than Sycamore. Willow also showcased its capability to solve complex problems in minutes that would be impractical for classical supercomputers. Together, these milestones reflect Google's commitment to advancing quantum performance, scalability, and real-world applications [55].

2.7.4 | Tradeoff of Quantum Computing Architectures

The quest for effective quantum processing has driven the creation of many quantum computing designs, each distinguished by special advantages and disadvantages that affect their fitness for certain computer applications. Although quantum computing has great promise, evaluating the real advantages of its uses is difficult as different methods frequently show sensitivity to complicated technical details on underlying primitives and their complexity [56]. Large-scale quantum computers need the creation of quantum versions of hardware, architectures, languages, compilers, and abstraction layers, thereby translating classical

ideas to quantum systems [57]. Various quantum computing systems show different degrees of development and present different difficulties with scalability, qubit quality, and coherence [26]. Two main modalities in quantum computing are trapped ions and superconducting circuits; each has special benefits and drawbacks [58, 59]. Artificially manufactured quantum circuits, or superconducting qubits, gain from well-known microfabrication methods, which allow for large qubit counts and good connections [60]. They struggle, meanwhile, with coherence, fidelity, and crosstalk [61]. Trapped ion qubits with high-fidelity operations and remarkable coherence times use individual ions constrained by electromagnetic fields. But the complexity of ion trapping and management usually limits scalability and connection. Additionally, under investigation as potential qubit candidates are neutral atoms and photons, each with unique benefits in terms of coherence, connection, and scalability. Although quantum computing has advanced significantly recently, large error rates remain a serious obstacle to the solution of practical issues, even if this area has achieved great development [62]. In terms of scalability and error correction, many quantum architectures, including superconducting qubits and trapped ions, offer different benefits and drawbacks.

- **Superconducting Qubits:** Made using well-known microfabrication processes, they are solid-state electronics. Their somewhat modest size and simplicity of integration help them to usually provide decent scalability. Higher mistake rates might result from their increased sensitivity to noise and decoherence, however.
- **Trapped Ions:** Individual ions caught in situ by electromagnetic fields are known as trapped ions. Compared to superconducting qubits, they usually have lower error rates and longer coherence periods. But the difficulties of capturing and regulating so many ions make scaling them more challenging. The particular use and the intended trade-offs between scalability and error correction determine the architectural option. Trapped ions might be favored for applications needing great precision; superconducting qubits may be more appropriate for applications requiring many qubits.

Many technological platforms, especially those created from superconducting devices, are beginning to exhibit similar tendencies as researchers hunt for designs suited for quantum computing scaling [63]. Recent industry roadmaps begin to suggest a modular method to scale these devices [63]. Aiming to use the complementary qualities of several qubit modalities, the research on hybrid quantum architectures is gathering impetus.

2.7.5 | Hybrid Quantum-Classical Heterogeneous Computing

Tightly integrated hybrid quantum-classical systems are increasingly important for pragmatic uses as quantum technologies develop. These systems run computations effectively by combining traditional high-performance computing (HPC) nodes with quantum processing units (QPUs) [64]. Ultra-fast feedback loops (sub-4 μ s), required for real-time quantum error correction and parameter optimization, are made possible via hardware-level interfaces such as NVIDIA DGX Quantum and FPGA-QPUs.

Early implementations by combining QPUs within conventional supercomputer stacks have shown advances in simulations of highly correlated materials, automated control systems, and LLM fine-tuning via quantum-enhanced layers [65]. These findings show that hybrid systems are not only feasible but also sensible in the NISQ era, as they provide the most likely way to get near-term quantum advantage across many fields.

2.8 | Fault-Tolerant Quantum Computing and Quantum Error Correction

In this section, the difference between quantum fault-tolerant and error correction is given in detail, along with the detailed strategies that are carried out in recent years.

2.8.1 | Fault-Tolerant Quantum Computing

Fault tolerance is a cornerstone in the development of practical quantum computing, addressing the inherent challenges posed by decoherence and environmental noise. Quantum systems are highly susceptible to errors due to their fragile nature, making fault tolerance a critical requirement for reliable quantum computation. Quantum Error Correction (QEC) plays a pivotal role in achieving fault tolerance by encoding logical qubits across multiple physical qubits, thereby mitigating errors arising from noise, gate imperfections, and measurement inaccuracies [66, 67]. Recent advancements in QEC have demonstrated significant progress in both theoretical frameworks and practical implementations, bringing fault-tolerant quantum computing closer to reality [68]. Fault tolerance is a critical milestone in quantum computing, addressing challenges from decoherence and environmental noise. Recent breakthroughs highlight practical implementations of error-corrected systems, particularly through the development of logical qubits and scalable error correction techniques [69]. Key advancements include:

1. **Logical Qubits:** Logical qubits are formed by encoding single logical states into multiple physical qubits using error correction codes. Notable progress in this area includes [70, 71]:
 - **Surface Codes:** Topologically, quantum error-correcting codes with large error thresholds and scalability are called surface codes. By embedding logical qubits into a lattice of physical qubits, they guard logical qubits and enable error detection and correction free from direct qubit measurement [72].
 - **48 Logical Qubit Demonstrations:** Recent demonstrations combining many logical qubits, notably Google's use of surface codes on a 5×5 qubit lattice, reveal development in error correction methods fundamental for fault-tolerant quantum computing.
2. **Scalable Error Correction:** Continuous syndrome measurements enable scalable error correction to find and fix errors in real-time. Lattice surgery and code concatenation are used to control error in large quantum systems, including [66, 73]:
 - **Shor's Algorithm:** Improved ability to factor large numbers, a critical step toward breaking classical cryptographic systems.

- HHL Algorithm: More accurate solutions for linear systems, with applications in optimization and machine learning.

2.8.2 | Core Concepts of Quantum Error Correction

Quantum systems are fundamentally different from classical systems in their error behavior. Unlike classical bits, quantum bits (qubits) cannot be directly measured or copied without collapsing their state, a limitation encapsulated by the no-cloning theorem. QEC overcomes this challenge by encoding logical qubits into a higher-dimensional Hilbert space using redundancy [74]. This encoding allows for the detection and correction of errors without directly measuring the quantum state. The QEC process involves three key steps:

- **Encoding:** Logical qubits are mapped to a larger set of physical qubits using error-correcting codes.
- **Syndrome Measurement:** Errors are detected through stabilizer operators or error syndromes, which provide information about the type and location of errors without collapsing the quantum state.
- **Error Correction:** Specific operations are applied based on the syndrome to correct the detected errors. QEC addresses various types of errors, including bit-flip errors (X errors), phase-flip errors (Z errors), and depolarizing errors, which randomize a qubit's state. By localizing and correcting these errors, QEC ensures the integrity of quantum computations. Recent progress in QEC spans advancements in code design, integration with emerging hardware, and interdisciplinary applications. Key developments include:
 - **High-Performance Codes:** Advanced codes like quantum low-density parity-check (qLDPC) and many-hypercube codes improve error tolerance and scalability.
 - **Machine Learning in Decoding:** Adaptive, data-driven approaches, such as neural network decoders, enhance the ability to correct errors in complex noise environments.
 - **Hybrid Strategies:** Combining classical and quantum error correction balances resource efficiency with fault tolerance, addressing unique challenges in hybrid architectures.

Despite advancements, QEC faces major challenges:

- **Qubit Overhead:** High physical qubit needs hinder logical qubit encoding.
- **Decoding Complexity:** Managing large systems with varied errors is computationally demanding.
- **Integration Challenges:** Adapting QEC to various quantum architectures like superconducting, photonic, and trapped-ion platforms requires custom solutions.

Emerging trends in QEC include topological qubits, distributed quantum networks, and quantum-assisted climate modeling, showing potential societal and environmental benefits [75]. Future research will focus on interdisciplinary collaboration, scalable architectures, and resource-efficient error correction.

QEC is not just technical but foundational to the quantum revolution, ensuring reliable computations for robust quantum technologies. Progress in QEC is key to realizing quantum computing's potential in cryptography, material science, healthcare, and energy. Interdisciplinary research and practical implementations are crucial to overcoming current challenges and achieving quantum technology's full promise.

2.9 | Quantum Programming Languages

The emergence of quantum computing marks a revolutionary shift in the landscape of technology, prompting the creation of specialized programming languages and libraries tailored for the unique challenges and capabilities of quantum mechanics. Unlike classical computing, which relies on bits as the fundamental unit of information, quantum computing utilizes qubits, enabling a fundamentally different approach to processing and manipulating data [76]. This paradigm shift has spurred the development of various quantum programming languages, each designed to facilitate the simulation, implementation, and optimization of quantum gates and circuits. These quantum programming languages serve as essential tools for researchers, developers, and enthusiasts eager to delve into the intricate world of quantum algorithms and protocols [77]. They provide a robust framework for experimenting with quantum phenomena, enabling users to harness the power of quantum superposition and entanglement to solve complex problems that are intractable for classical computers [78]. In this section, we will explore some of the most prominent programming languages utilized for quantum simulations. We will delve into their distinctive features, advantages, and use cases, highlighting how each language caters to the specific needs of quantum computing [79]. By understanding these languages, we can better appreciate the vast potential of quantum computing and its implications for various fields, including cryptography, optimization, and artificial intelligence [80].

2.9.1 | Q# (Q Sharp)

Q# is a high-level, open-source programming language developed by Microsoft for quantum computing. It is part of the Quantum Development Kit (QDK) and is designed to facilitate the development and execution of quantum algorithms. One of the key features of Q# is its hardware-agnostic nature, meaning that qubits in quantum algorithms are not tied to specific quantum hardware or layouts. This allows for greater flexibility and compatibility with various quantum processors [81].

Q# integrates both quantum and classical computing, enabling the execution of classical computations alongside quantum operations. This hybrid approach is essential for the development of practical quantum applications. The language respects the fundamental principles of quantum mechanics, ensuring that quantum algorithms adhere to the laws of physics. For instance, Q# prevents direct copying or accessing of qubit states, maintaining the integrity of quantum information.

The structure of a Q# program typically includes user-defined namespaces, operations, and functions. Operations are the

primary building blocks, representing quantum subroutines that manipulate qubit states. Q# also provides a rich standard library with predefined operations and functions, such as the Hadamard operation (H) for creating superposition states and the Measure operation for qubit measurement.

Q# supports various development environments, including Visual Studio Code and Jupyter Notebooks, and can be integrated with Python for enhanced functionality. The Quantum Development Kit also includes simulators and resource estimators, allowing developers to test and optimize their quantum algorithms before running them on actual quantum hardware. This comprehensive toolset makes Q# a powerful language for exploring and advancing the field of quantum computing.

2.9.2 | Qiskit (Quantum Information Science Kit)

Qiskit is an open-source software development kit (SDK) created by IBM for quantum computing. It provides a comprehensive set of tools for building and executing quantum algorithms on quantum computers and simulators. The core component of Qiskit is its SDK, which includes modules for creating and manipulating quantum circuits, operators, and primitives [82].

One of the standout features of Qiskit is its circuit-building tools, which allow users to initialize and manipulate quantum registers, circuits, instructions, gates, parameters, and control flow objects. The circuit library offers a wide range of pre-built circuits, instructions, and gates, essential for circuit-based quantum computations. Additionally, the quantum information library provides a toolkit for working with quantum states, operators, and channels, enabling exact calculations without sampling noise [83].

Qiskit also includes a transpiler, which optimizes quantum circuits for execution on specific quantum processing units (QPUs). The primitives module contains base definitions and reference implementations of the Sampler and Estimator primitives, which can be adapted by different quantum hardware providers.

Qiskit Runtime is a cloud-based service that streamlines quantum computations on IBM Quantum hardware, offering optimal implementations of Qiskit primitives. It supports error suppression and mitigation techniques to enhance the quality of quantum circuit execution.

Moreover, Qiskit Serverless provides a simple interface for running workloads across quantum-classical resources, facilitating the deployment of programs to the IBM Quantum Platform and managing resources for multi-cloud and quantum-centric supercomputing use cases.

Overall, Qiskit is a powerful and versatile tool for quantum computing, supporting a wide range of applications and research in the field.

2.9.3 | Cirq (Circuit Language for Near-Term Quantum Computation)

Cirq is an open-source quantum computing framework developed by Google Quantum AI, designed to facilitate the creation,

simulation, and optimization of quantum circuits. It provides a comprehensive set of tools for researchers and developers to build quantum algorithms and run them on both simulators and actual quantum hardware [84].

At its core, Cirq allows users to define qubits, which can be labeled in various ways, such as “NamedQubit”, “LineQubit”, and “GridQubit”. These qubits serve as the fundamental units for constructing quantum circuits. Cirq supports a wide range of quantum gates and operations, including single-qubit gates like the Hadamard (H) gate and multi-qubit gates like the Controlled-NOT (CNOT) gate. Users can create complex quantum circuits by combining these gates and operations.

Cirq also emphasizes the importance of device compatibility, providing tools to ensure that circuits adhere to the constraints of specific quantum hardware. This includes validating circuits against device-specific rules and optimizing them for efficient execution.

Simulation is a key feature of Cirq, enabling users to test and debug their quantum circuits before deploying them on actual quantum processors. Cirq’s simulator can handle circuits with up to 20 qubits, offering both state vector simulation and sampling-based approaches to mimic real quantum hardware behavior.

Additionally, Cirq supports parameterized circuits and parameter sweeps, which are essential for variational algorithms and optimization tasks. The framework also includes tools for visualizing results, such as plotting state histograms and analyzing measurement outcomes.

Overall, Cirq provides a robust and flexible platform for advancing quantum computing research and development, making it accessible to both beginners and experienced practitioners.

PennyLane is an open-source Python library designed for programming quantum computers, developed by Xanadu. It facilitates the integration of quantum computing with machine learning frameworks such as NumPy, JAX, PyTorch, and TensorFlow, enabling the creation of quantum-aware applications. PennyLane’s differentiable programming paradigm allows for the execution and training of quantum programs on various backends, including both simulators and hardware devices [85].

A key feature of PennyLane is its ability to manage the execution of quantum computations, including the evaluation of circuits and the computation of their gradients. This capability is crucial for creating seamless quantum-classical pipelines, which are essential for applications in quantum machine learning and quantum chemistry. PennyLane’s design ensures that circuits can be run on different types of simulators or hardware without modification, optimizing communication with devices and compiling circuits to suit the backend [86].

The library comes with default simulator devices and is well-integrated with external software and hardware platforms such as IBM’s Qiskit and Google’s Cirq. This integration allows

TABLE 1 | Key quantum algorithms: Applications, complexity & feasibility.

Algorithm	Applications	Complexity	NISQ Feasibility/Status
Variational Quantum Eigensolver (VQE)	Chemistry: Ground-state energy, materials simulation	Polynomial(n)	High/shallow circuits, error resilience [87]
Quantum Approximate Optimization Algorithm (QAOA)	Combinatorial optimization (Max-Cut, scheduling)	Polynomial(n)	High/Heuristic, parameter tuning on NISQ [88]
Grover's Search	Unstructured search, NP class problems	$O(\sqrt{N})$ queries	Moderate/Requires coherent oracle and amplitude amplification [18]
Shor's Algorithm	Integer factorization, discrete logarithms	Polynomial(log N)	Low/Needs fully fault-tolerant QEC and thousands of qubits [17]
Deutsch-Jozsa	Black-box function classification (constant vs. balanced)	$O(1)$	Demonstration/Shallow, for theoretical and educational purpose [89]
Quantum Walk Search	Graph search, element distinctness	Quadratic speed-up: $O(\sqrt{N})$	Research-stage/Useful in graph-based tasks
Quantum Phase Estimation (QPE)	Eigenvalue estimation, HHL, cryptography	Polynomial(log precision) deep circuits	Low/Heavy coherence demands, fault-tolerant needed [90]

users to leverage a wide range of quantum computing resources for their research and development needs.

PennyLane's versatility and robust feature set make it a valuable tool for researchers and developers in the field of quantum computing. Its ability to bridge the gap between quantum and classical computing frameworks positions it as a leading platform for advancing quantum technologies and their applications.

2.10 | Evolution of Quantum Algorithms

The remarkable progression of quantum computing, which has the capacity to revolutionize various fields and industries, can be attributed predominantly to the groundbreaking advancements made in the realm of quantum algorithms depicted in Table 1, which skillfully capitalize on the distinctive and often counter-intuitive characteristics exhibited by qubits to attain significant computational benefits that far surpass those offered by traditional classical computing methodologies. In stark contrast to classical algorithms, which are inherently limited by the linear and sequential nature of processing classical bits, quantum algorithms harness the extraordinary phenomenon of quantum parallelism, thereby enabling them to tackle intricate and multifaceted problems with a level of efficiency and speed that is particularly advantageous in designated areas of application.

2.10.1 | Deutsch-Jozsa Algorithm

It was the first quantum algorithm to demonstrate exponential speedup over classical methods. It solves a specific black-box problem—determining whether a function is constant or balanced—with a single quantum query, whereas classical methods require at least two queries. This algorithm provided the first concrete proof that quantum computers could outperform classical computers for certain tasks, marking the beginning of quantum algorithm research. While primarily theoretical, it laid the foundation for understanding quantum parallelism and interference, which are essential for more advanced algorithms. Its

significance lies in its role as a stepping stone for the development of more complex quantum algorithms [89].

2.10.2 | Quantum Fourier Transform (QFT)

QFT is a quantum analogue of the classical Fourier Transform and is a key component of Shor's algorithm. It enables efficient solutions to problems like integer factorization and discrete logarithms by leveraging the principles of quantum superposition and interference. QFT is a cornerstone of quantum computing, enabling exponential speedups for problems that rely on periodicity or frequency analysis. Its applications span cryptography, signal processing, and quantum simulations, making it one of the most fundamental tools in quantum algorithm design [91].

2.10.3 | Shor's Algorithm

This algorithm was developed by Peter Shor, is a groundbreaking algorithm that factors large integers and computes discrete logarithms exponentially faster than classical methods. By leveraging QFT and modular arithmetic, Shor's algorithm threatens the security of widely used cryptographic systems like RSA and ECC. Its implications have driven global efforts to develop quantum-resistant cryptography and quantum key distribution (QKD) protocols. Shor's algorithm is one of the most influential quantum algorithms, as it demonstrates the potential of quantum computing to disrupt classical cryptographic systems and has spurred significant advancements in cybersecurity [17].

2.10.4 | Grover's Algorithm

This algorithm was developed by Lov Grover, provides a quadratic speedup for unstructured search problems. It can search an unsorted database of N items in $O(\sqrt{N})$ time, compared to $O(N)$ for classical methods. While not as dramatic as exponential speedups, Grover's algorithm is widely applicable and has practical implications for optimization, database searching, and artificial intelligence. Its applications include database

searching, optimization problems, machine learning, and cryptography (e.g., brute-force attacks). Grover's algorithm is particularly valuable for its versatility and relevance to real-world problems [18].

2.10.5 | Harrow-Hassidim-Lloyd (HHL) Algorithm

The algorithm is designed to solve linear systems of equations exponentially faster than classical methods, provided the matrix is sparse and well-conditioned. It is a quantum analogue of classical linear algebra techniques and has broad implications for scientific computing, as solving linear systems is a fundamental task in many fields. Applications of the HHL algorithm include machine learning (e.g., support vector machines, regression), electromagnetics, fluid dynamics, and financial modeling. Its ability to accelerate computations central to scientific and engineering tasks makes it a powerful tool in the quantum computing toolkit [92].

2.10.6 | Quantum Phase Estimation (QPE)

QPE is a quantum algorithm used to estimate the eigenvalues of a unitary operator. It is a key subroutine in many quantum algorithms, including Shor's algorithm and quantum simulations. QPE enables the extraction of information about quantum systems, making it essential for quantum chemistry, material science, and cryptography. Its applications include quantum simulations, eigenvalue problems, and quantum chemistry. QPE is a fundamental building block for many advanced quantum algorithms and plays a critical role in understanding and manipulating quantum systems [90].

2.10.7 | Variational Quantum Algorithms (VQAs)

VQAs are hybrid quantum-classical algorithms that use quantum computers to evaluate cost functions and classical computers to optimize parameters. Examples include the Variational Quantum Eigensolver (VQE) and Quantum Approximate Optimization Algorithm (QAOA). VQAs are designed for near-term quantum devices (NISQ era) and are robust to noise, making them practical for current hardware. Their applications span quantum chemistry, optimization, machine learning, and material science. VQAs represent a significant step toward practical quantum computing by enabling the use of imperfect quantum hardware for real-world problems [87].

2.10.8 | Quantum Machine Learning (QML)

QML Algorithms leverage quantum computing to enhance classical machine learning tasks. Examples include quantum support vector machines (QSVM), quantum neural networks (QNN), and quantum principal component analysis (QPCA). Quantum machine learning (QML) has the potential to accelerate tasks like classification, clustering, and dimensionality reduction, particularly for large datasets. Applications of QML include artificial intelligence, data analysis, and pattern recognition. As quantum hardware improves, QML algorithms are expected to play an important role in the field of machine learning [93].

2.10.9 | Variational versus Pure Quantum Algorithms

The modern NISQ era clearly distinguishes hybrid variational methods from pure quantum methods. Combining brief, parameterized quantum circuits with conventional optimizers in a feedback loop, VQAs such as the Variational Quantum Eigensolver (VQE) and QAOA work together. These designs are specifically designed for noisy hardware; they withstand faults by maintaining modest circuit depths and offloading problems to conventional CPUs [94]. For near-term quantum advantage, VQAs have consequently become the most exciting techniques. Conversely, pure quantum algorithms—such as Shor's algorithm or QPE—rely totally on deep, error-corrected quantum circuits running in a fault-tolerant environment. On present NISQ devices, they are impractical as they need extended coherence durations and complete quantum error correction.

3 | Quantum Computing Applications

Quantum computing applications are vast and diverse, with the potential to revolutionize many fields by solving problems beyond classical methods. From cryptography and optimization to drug discovery and materials science, quantum computing can transform industries and drive innovation. This review study delves into the practical applications of quantum computing, offering insights into how this nascent technology can be leveraged to tackle real-world challenges and unlock novel opportunities for scientific and technological progress.

3.1 | Cryptography

Quantum computing signifies a transformative advancement in computational capabilities, enabling the exploitation of quantum mechanical principles to execute complex calculations at unparalleled velocities. This groundbreaking technology holds substantial potential across various domains, notably in cybersecurity. As cyber threats evolve in sophistication, conventional cryptographic techniques may prove inadequate for safeguarding sensitive information. QC introduces methods to enhance cybersecurity, including quantum-resistant encryption and improved threat detection. Leveraging quantum computing can secure digital infrastructures against emerging threats, ensuring data integrity and confidentiality.

Chuck Brooks, a preeminent expert in cybersecurity and the chair of the Quantum Security Alliance, emphasizes the critical necessity for robust security measures through collaborative efforts among academic institutions, industry stakeholders, researchers, and governmental bodies [95]. The security of quantum computing systems is of utmost importance, given their potential for exponential computational advancements that could undermine existing cryptographic protocols. While cryptography has traditionally been the cornerstone of healthcare information security, the integration of quantum computing with cryptography merges the principles of classical cryptography and quantum mechanics. This amalgamation ensures unconditional security for healthcare communications among service consumers. Quantum cryptography, as the first commercially viable application of quantum computing, signifies a pivotal advancement in achieving superior security.

3.1.1 | Quantum Key Distribution

Quantum Key Distribution (QKD) is a sophisticated protocol devised to authorize two parties by securely distributing a mutually agreed-upon key for communication. Leveraging fundamental quantum laws, often rooted in the intricate principles of quantum computing, the QKD protocol excels in detecting information extraction attacks. It exploits the distinctive signatures left by adversaries attempting to intercept information, thereby facilitating the detection of potential security breaches. QKD supports the generation of keys of arbitrary length and automatically ceases the key generation process upon detecting an attack. The seminal QKD technique, BB84, introduced by Gilles Brassard, is widely used in quantum computing research [96]. QKD addresses key management and distribution challenges in classical algorithms. Shor et al. validated BB84's security, linking it to entanglement purification and quantum error correction [97]. Extensive research on QKD security protocols has led to significant advancements in quantum computing security, enhancing sensitive data transmission security across various applications.

3.1.2 | Quantum D-Level Systems

Quantum D-level systems are advantageous for many applications, especially due to their superior data transfer rates, relevant for next-generation sensor technologies. In Quantum Key Distribution (QKD), d-level protocols significantly enhance key transmission rates and error resistance.

In Reference [98], d-level systems were used to protect against individual and concurrent attacks. Two cryptosystems were discussed: One with two mutually unbiased bases and another with $d+1$ unbiased bases. Security proofs for protocols involving entangled photons against individual attacks noted increased error rates [99]. The decoy pulse method for BB84 in high-loss scenarios replaces signal pulses with multiphoton pulses [100]. Security proofs for coherent-state protocols using Gaussian-modulated coherent states and homodyne detection against arbitrary coherent attacks were presented [101]. Another study proposed security measures against common attacks on quantum channels by powerful eavesdroppers [102]. Device-independent (DI) QKD security against collective attacks was analyzed in Reference [103].

A passive security approach using a beam divider to segregate each input pulse was detailed in Reference [104]. These developments highlight ongoing efforts to improve the security and efficiency of quantum communication protocols.

3.1.3 | Finite Key Analysis Method

The finite key analysis method is prominent in Quantum Key Distribution (QKD) and is part of composable unconditional security proofs. It addresses security constraints related to finite-length keys in practical BB84 environments, including prepare and measure implementations without decoy states and entanglement-based techniques [105]. Song et al. outline finite-key analysis of Measurement-Device-Independent (MDI) QKD, eliminating major detector channels and introducing

key-rate schemes surpassing full-device-independent QKD [106]. Curty et al. provide a security proof against general attacks in finite-key, demonstrating long-distance MDI QKD feasibility within specific transmission time-frames [107]. Zhou et al. present a partial device-independent BB84 protocol constrained by finite resources [108]. Wang et al. detail security analysis against communication exposure during preparation [109]. These studies highlight efforts to ensure robust QKD security in practical scenarios with finite key lengths and resource constraints.

3.1.4 | Device-Independent Quantum Key Distribution

Device-Independent Quantum Key Distribution (DI QKD) seeks to address the practical implementation gap in Quantum Key Distribution without delving into the working mechanisms of the underlying quantum devices [103]. It relies on the violation of Bell's inequalities between both communication endpoints, offering increased security compared to classical schemes by minimizing security assumptions. An alternative approach involves information receivers at both ends identifying the infringement of the Bell inequality. The essence of DI-QKD lies in not needing information about the underlying devices, which may include potential adversaries. Consequently, the emphasis is on identifying elements rather than specifying how quantum security is implemented [110]. In security vulnerabilities, DI-QKD is shown to be capable of defending against various types of attacks, including phase-remapping attacks [111], binding attacks [112], and wavelength-dependent attacks [113]. Furthermore, the technique proposed in Reference [114] addresses the identification and defense of security vulnerabilities arising from quantum communication channels. Broadbent et al. introduced a generalized DI QKD protocol based on two-mode Schrödinger cat states [115]. In summary, DI QKD stands as a versatile approach capable of providing robust defense against a range of security vulnerabilities in quantum communication channels.

3.1.5 | Semi-Quantum Key Distribution (SQKD)

Semi-Quantum Key Distribution (SQKD) leverages the quantum capabilities of at least one party to reduce computational overhead and costs, achieving Quantum Key Distribution (QKD). Only the sender needs quantum capabilities, while the receiver can use classical capabilities. The sender prepares, measures, and stores quantum states, while the receiver prepares, measures, arranges, and transmits qubits without disturbing the quantum channels. Boyer et al. proposed the first SQKD protocol in 2007 using single-photons, later extending it to prove robustness when qubits are individually transmitted but collectively attacked [116, 117]. Zou et al. introduced a robust protocol with fewer than four quantum states [118]. Maitra et al. analyzed a two-way eavesdropping scheme [119], and Krawec et al. proposed a secret key-sharing scheme for classical users [120]. Zou et al. [121] addressed the sender's measurement capabilities, ensuring robustness against joint attacks, and noted that classical users' measurement capability is not essential. Liu et al. explored an SQKD protocol with an untrusted quantum server attempting to steal session keys [122]. Various quantum states

and technologies are being used to develop new SQKD protocols [123, 124], highlighting the evolving landscape of SQKD and its secure communication benefits while minimizing computational complexities.

3.2 | Medicine

Quantum computers possess the innovative potential to revolutionize diverse domains, notably the diagnosis of diseases and drug discovery. Leveraging the principles of quantum mechanics, these sophisticated computational systems are capable of processing and analyzing vast datasets with unparalleled speed and efficiency. This capability is particularly advantageous in the medical field, where early and precise diagnosis is imperative for effective therapeutic interventions. Quantum computers can augment the accuracy of diagnostic tools, facilitate the identification of novel biomarkers, and enhance the efficiency of intricate simulations. Furthermore, they can markedly expedite the drug discovery process by simulating molecular interactions and identifying promising drug candidates with greater efficacy. Ultimately, this technological advancement contributes to improved patient outcomes and significant progress in personalized medicine [125].

3.2.1 | Drug Discovery

The process of drug discovery is intricate and long-lasting, encompassing the identification of potential new drugs and their introduction to the market. This typically involves several stages: Target identification, lead compound discovery, preclinical testing, clinical trials, and regulatory approval. Traditional methods often depend on trial and error, which can be both time-consuming and expensive. Quantum computers have the potential to transform drug discovery by executing complex calculations much faster than classical computers. These advanced machines utilize the principles of quantum mechanics to solve problems that are currently unsolvable by classical systems.

In the realm of drug discovery, quantum computers can simulate molecular structures and interactions with unprecedented detail, enabling researchers to identify promising drug candidates more efficiently. By speeding up the computational aspects of drug discovery, quantum computers could significantly reduce the time and cost associated with bringing new drugs to market. Quantum computing allows medical professionals to model molecular interactions at the atomic level, a crucial aspect of medical research [126]. This capability is particularly significant for various medical applications, including diagnosis, treatment, drug discovery, and analytics. Recent advances in quantum computing now enable the encoding of tens of thousands of proteins and the simulation of their interactions with drugs, a feat previously unattainable [127]. The processing of such complex information is significantly more efficient, surpassing conventional computing capabilities by orders of magnitude [128]. Quantum computing facilitates the simultaneous comparison of vast datasets and their permutations, aiding in the identification of optimal patterns. With the use of gold nanoparticles and established methods like the bio-barcode assay, it is now possible to detect disease-specific biomarkers in blood [129]. The goal is to

leverage these comparisons to facilitate accurate identification of the diagnosis.

QC revolutionizes medical research by analyzing molecular interactions at the atomic level. Quantum computers can model small molecules like lithium hydride [130] and excel in quantum chemistry calculations that require explicit wave function representation, benefiting from high entanglement and accurate property simulations [131]. Quantum algorithms for chemistry, such as those estimating ground-state energies and reaction rates, outperform classical methods. Huggins et al. demonstrated quantum computing's accuracy in noisy circuitry simulations using up to 16 qubits on Google's 53-qubit quantum computer for Monte Carlo simulations of fermion models, including complex molecules like H₄, molecular nitrogen, and solid diamond, involving up to 120 electron orbitals [132]. This highlights quantum computers' potential in complex quantum chemistry simulations with real-world applications.

3.2.2 | Precision Medicine

Precision medicine addresses individual healthcare needs, recognizing the complexity of the human biological system beyond traditional treatments. Classical machine learning (ML) has been effective in predicting disease risks using electronic health records (EHRs), but issues like data quality, noise, feature size, and complex feature relationships persist [133]. This has led to exploring quantum-enhanced ML for more accurate early disease detection [134].

Healthcare professionals can use quantum-enhanced ML tools for continuous virtual diagnosis based on ongoing data streams. Research in drug sensitivity at the cellular level, considering cancer cells' genomic characteristics, aims to predict cancer treatment efficacy. Quantum-enhanced ML shows promise in accelerating healthcare breakthroughs through drug inference models [135, 136].

The goal of precision medicine is to clarify the relationships between causes and treatments, predicting personalized actions. Traditional diagnosis based on symptoms can lead to umbrella diagnoses, where treatments may fail [137]. Quantum computing can use continuous data streams for personalized interventions, predicting diseases, and facilitating treatments. Quantum-enhanced predictive medicine personalizes healthcare through continuous care, potentially improving patient adherence and participation [138].

3.3 | Financial Modeling

Quantum computing is enhancing the efficiency of multi-asset portfolio simulations. The research conducted by Sun et al. utilizes IBM's quantum computing platform to facilitate financial applications, thus laying a foundational software framework for prospective quantum applications within the finance sector [139]. The efficacy of the developed algorithm is evaluated through a multi-period asset allocation problem. Quantum computing is poised to revolutionize financial risk management by providing sophisticated tools for risk analysis and optimization, ultimately conferring a competitive advantage.

The study by Adegbola et al. delves into the ramifications of quantum computing on financial risk management, offering actionable insights on how to incorporate quantum solutions into established frameworks. Quantum computing holds potential for the banking industry, particularly through quantum machine learning that augments financial modeling and analytical capabilities [140]. This literature review by Bunescu and Vârtei comprehensively examines the applications of quantum computing in finance, emphasizing the areas of simulation, optimization, and machine learning. It systematically highlights the advantages and diverse applications of quantum computing within the financial domain [141]. Moreover, quantum computing significantly enhances financial portfolio optimization by employing advanced quantum algorithms. Notably, quantum annealing and the Quantum Approximate Optimization Algorithm (QAOA) are explored as methodologies for addressing challenges in portfolio optimization. Owolabi et al. investigate the effectiveness of these quantum algorithms, assessing their potential to deliver computational speedup in comparison to classical approaches [88].

In the realm of derivative pricing, which entails martingale probability measures and market models, quantum techniques are employed to efficiently tackle pricing challenges within incomplete markets. The framework for these pricing problems is articulated through linear algebraic formulation, utilizing quantum methods to yield efficient solutions for derivative pricing as discussed by Rebentrost et al. The advancements in quantum computing present an opportunity for the finance sector, particularly in areas such as multi-asset portfolio simulations, financial risk management, and derivative pricing [142]. The studies highlight the implementation of advanced quantum algorithms like quantum annealing and the Quantum Approximate Optimization Algorithm (QAOA), demonstrating their potential to significantly enhance computational efficiency over classical methods. As the literature meticulously shows, these innovations are not just theoretical; they pave the way for practical applications that could redefine financial modeling and analytical capabilities, thereby providing a competitive edge in an increasingly complex financial landscape. The integration of quantum solutions into existing frameworks is poised to enable more sophisticated tools for risk analysis and optimization, marking a pivotal shift in how the finance industry operates.

3.4 | Optimization Problems

Quantum computing represents a groundbreaking advancement with the potential to revolutionize the way we approach optimization problems across a multitude of disciplines. By harnessing its distinctive computational capabilities, quantum computing offers innovative solutions that traditional computing methods struggle to achieve. A particularly noteworthy strategy is the deployment of quantum optimization algorithms that utilize multistep quantum computation. This approach effectively reduces the dimensionality of the search space, allowing for a more streamlined process in identifying optimal solutions. By confining the state space, this method significantly alleviates challenges related to local minima, which often complicate the search for the best possible outcome in complex optimization scenarios [143].

The ability of quantum computing to tackle NP-hard optimization problems—a class of issues that are notoriously difficult and prevalent in industries such as logistics, finance, and supply chain management—has garnered considerable interest from researchers and practitioners alike. Techniques such as quantum annealing and gate-based quantum computing have emerged as frontrunners in this field. For instance, the Quantum Approximate Optimization Algorithm (QAOA) has shown remarkable potential in addressing intricate optimization challenges, delivering solutions with improved efficiency and accuracy compared to classical counterparts [144].

Furthermore, the investigation into quantum algorithms tailored for engineering optimization, including Grover's search algorithm and quantum phase estimation, highlights the capacity for extensive parallelism in solution searching. This characteristic is particularly beneficial in specialized fields such as materials design and topology optimization, where the complexity of the problems often demands innovative and efficient approaches [145].

In summary, quantum computing offers substantial advantages in the realm of optimization problems, providing not only efficient and scalable solutions but also the promise of transformative impacts across various sectors. However, to fully harness its capabilities and integrate quantum solutions into practical applications, ongoing research and development efforts are essential. As the field continues to evolve, the potential for quantum computing to redefine optimization paradigms remains an exciting frontier in technology and science.

3.5 | Quantum Machine Learning

Recent literature in the realm of quantum computing as applied to machine learning and artificial intelligence showcases a variety of breakthroughs, algorithms, and use cases showing the potential of quantum-enhanced technologies to revolutionize these fields [146].

Quantum machine learning utilizes the unique capabilities of quantum computing, such as parallel computation and quantum entanglement, to enhance traditional algorithms and tackle complex problems efficiently. Studies have outlined key quantum algorithms like Quantum Support Vector Machines (QSVM), Quantum Principal Component Analysis (QPCA), and Quantum K-Nearest Neighbors (Q-KNN), emphasizing their advantages in processing power and complex problem-solving in AI and machine learning applications [93]. Specific advancements are noted in image classification, where quantum machine learning methods are demonstrated to offer higher efficiency, accuracy, and faster processing of complex data through simulations and hybrid quantum-classical neural network models [147].

Moreover, quantum machine learning integrates quantum technology with artificial intelligence to employ quantum-enhanced algorithms and hybrid quantum-classical neural networks for improving model accuracy and reducing computational demands [148]. The field, being in its early stages, shows promise for faster learning from data, handling larger or more complex datasets, and the potential for significant impacts across various scientific

TABLE 2 | Future roadmap of current engineering and healthcare applications of quantum computing.

Focus area	Current applications	Near-Term (1–3 years)	Mid-Term (4–7 years)	References
Medical Imaging & AI	QCNN-based MRI, hybrid tumor detection pipelines	Clinical validation of quantum-enhanced tomography	Integration into radiology workflows; regulatory approval	[23, 25, 154]
Therapy Planning	VQE/QAOA optimization for treatment regimens	Hybrid pipelines tailored for personalized radiotherapy	Adaptive treatment planning with quantum feedback loops	[23, 25]
Engineering Simulations	Quantum-enabled CFD and structural analysis	Prototypical aerospace/civil simulations	Certified quantum-assisted CAD/simulation platforms	[155, 156]
Quantum Software Engineering	Hybrid QC-classical development tools	Full-stack QSE DevOps cycles	Integrated QSE pipelines for production systems	[157, 158]
Fault Tolerance & Scaling	Near-term NISQ prototypes; IBM's roadmap to 200 LQs	Quantinuum fault-tolerant "Apollo" systems by 2030	Universal fault-tolerant devices post-2030	[159, 160]

and technological fields if its advantages over classical algorithms are fully realized [149]. Quantum neural networks, in particular, significantly enhance machine learning capabilities by overcoming limitations of classical computational models through universal quantum computations, relevant for solving practical problems [150].

Further, supervised machine learning modeling with quantum computing has been addressed, focusing on feature selection and parameter encoding. This leverages quantum mechanisms like superposition and entanglement to enhance analysis and prediction on complex data tasks, thereby improving upon classical machine learning methods [151]. There's also an exploration of quantum computing's potential to enhance artificial intelligence and machine learning algorithms through increased speed of training and computational power, with AI aiding quantum computing in error correction algorithms. Successful implementations of unsupervised and supervised learning algorithms on quantum computers have been shown, reinforcing the symbiotic relationship between AI and quantum computing technologies [152].

The literature not only highlights the immediate advancements and applications but sets the stage for a future where quantum computing and AI might exponentially speed up learning and inference, offering groundbreaking improvements over classical systems and models [153]. The Table 2 shows the current engineering and healthcare applications of quantum computing.

3.6 | Materials Science

Materials science is undergoing a revolutionary transformation driven by the advent of quantum computing, which offers advanced methodologies for tackling intricate problems with a level of precision that far surpasses traditional classical computing techniques. One of the most notable methodologies emerging in this field is the Quantum Monte Carlo (QC-QMC) technique, particularly when it is integrated with hybrid tensor networks. This powerful combination facilitates highly accurate electronic structure calculations for extensive molecular and solid-state systems, allowing researchers to delve deeper into the complexities of material behavior at the quantum level.

The QC-QMC approach leverages sophisticated quantum algorithms, such as the variational quantum eigensolver (VQE), which is designed to yield exceptional accuracy while minimizing the quantum resources required for computation. This efficiency is crucial, as it enables scientists to explore larger systems and more complex interactions than previously possible, thereby accelerating the pace of discovery in materials science [161].

In addition to these advancements, the development of multi-scale quantum algorithms represents a significant leap forward in the simulation of complex material systems. By effectively merging classical and quantum computing techniques, these algorithms can efficiently model the intricate behaviors of materials, underscoring the importance of quantum computing not only in material design but also in the realm of drug discovery [162].

Moreover, practical applications of quantum computing in materials science extend to techniques such as quantum annealing, which is employed for configurational analysis of materials. This method provides computational chemists with a streamlined approach to explore the vast configurational space of materials, enabling quicker insights and facilitating the discovery of new compounds and materials [163].

Furthermore, ab initio quantum simulations utilizing quantum embedding techniques have emerged as a powerful tool for accurately modeling strongly correlated materials. These simulations have demonstrated their effectiveness in addressing challenges that are traditionally deemed intractable by classical methods, thus opening new avenues for innovation in industrial applications [164].

In summary, the integration of quantum computing into materials science not only enhances our understanding of material properties at a fundamental level but also significantly accelerates the discovery and design of new materials, with far-reaching implications for various scientific and industrial sectors.

3.7 | Analyzing the Practicality of Applied Quantum Algorithms on Modern Devices

Evaluating whether quantum algorithms might be implemented on present hardware requires a comprehensive knowledge of

the limits and developments in quantum computing technology. High error rates, few qubit counts, and the significant cost needed for error correction define the main difficulties.

3.7.1 | Error Rates and Qubit Stability

Physical qubit error rates of current quantum devices run approximately 0.1% per gate operation. Often demanding error rates below 0.01% per operation, such error rates are far greater than the thresholds needed for consistent execution of complicated quantum algorithms. This difference calls for fault-tolerant computing by means of Quantum Error Correction (QEC) methods.

3.7.2 | Quantum Error Correction Overhead

Using QEC results in significant overhead as a single logical qubit capable of withstanding mistakes requires many physical qubits. Depending on the error correction algorithm and the desired error suppression level, estimates point to hundreds to thousands of physical qubits needed per logical qubit. Using surface codes, for example, might need hundreds of physical qubits per logical qubit to get a logical error rate appropriate for pragmatic uses.

3.7.3 | Algorithmic Feasibility on Current Hardware

With the present constraints, only several quantum algorithms are practical on current systems:

- *Variational Quantum Algorithms (VQAs)*: Designed to operate within the confines of noisy intermediate-scale quantum (NISQ) devices, algorithms such as the Variational Quantum Eigensolver (VQE) and Quantum Approximate Optimization Algorithm (QAOA) are applied to issues in quantum chemistry and combinatorial optimization; they mix quantum circuits with conventional optimization.
- *Quantum Simulations*: With present qubit counts and coherence periods, it is possible to simulate quantum events or tiny quantum systems. But error accumulation and qubit constraints keep methods needing deep circuits and great fidelity, such as Shor's algorithm for factoring big integers, outreach of present hardware.

Recent advances offer promise in lowering the QEC resource needs. For instance, Microsoft's work with Quantinuum showed how sophisticated error correcting methods may create logical qubits from fewer physical qubits. Furthermore, suggesting development toward scalable, fault-tolerant quantum computing, Google's Willow processor achieved below-threshold quantum error correction. Notwithstanding these developments, the road to useful use of sophisticated quantum algorithms is still difficult. Reaching the required qubit counts, error rates, and coherence periods will call for ongoing creativity in quantum hardware design, error correction techniques, and algorithm development.

4 | Technical Challenges and Future Scope of Quantum Computing

While it has great potential to transform disciplines ranging from health to materials science, quantum computing must

overcome a great range of technological challenges before it can fully realize itself. Complicated chemical structures, supply networks, financial modeling, and risk assessment [165] are among the difficult challenges quantum computers provide answers for. By controlling the quantum states of qubits, one may accomplish intractable tasks for classical computers. Each of these issues—which cover qubit coherence, gate integrity, error mitigation, and scalability—presents special technical and scientific difficulty [166, 167]. Dealing with these issues will require coordinated efforts across many disciplines—including physics, computer science, materials science, and engineering [21, 57]. Currently limited to specialized data centers, quantum computers are constrained by qubits and quantum processor physical realization technologies [168]. Maintaining qubit coherence [62] is one of the most important difficulties in quantum computing. The basic building blocks of quantum computers, qubits, are naturally brittle and sensitive to external noise, which causes decoherence—the loss of quantum information [169]. Interactions between the qubits and their surroundings generate decoherence, which causes the qubits to lose their superposition and entanglement—the fundamental qualities that allow quantum computation [170]. Variations in temperature, electromagnetic radiation, and even stray atomic collisions may all contribute to this interaction. The difficulty is separating qubits from their surroundings to reduce these undesired interactions while still enabling controlled manipulation and measurement. Unlocking the most potent known uses of quantum computers requires the reduction of error rates of quantum devices [61]. Decoherence [171] is experienced by quantum systems, including optics. Realizing usable quantum computers also depends on reaching high-fidelity quantum gates. The basic operations that control qubits' states to execute computations are quantum gates; hence, their correctness directly affects the general integrity of a quantum system. Errors introduced by imperfect gate operations find their way into the calculation and may rapidly compound to make the outcomes useless. Factors like control faults, crosstalk between qubits, and gate physical implementation flaws restrict gate authenticity [172]. Furthermore, consistently producing qubits with enough performance marks is yet another major challenge for the creation of multi-qubit systems [26]. Improving gate authenticity calls for exact control over the physical parameters controlling qubit interactions as well as the development of sophisticated error-correction methods to minimize the consequences of gate faults [173]. Moreover, flaws in the coherent control of qubits might restrict the speed of a computer, as once relaxation periods are enough to enable fault-tolerant operation [174]. Achieving fault-tolerant quantum computing in quantum computers is severely challenged by error rates. Quantum systems are inherently noisy from decoherence, control flaws, and other kinds of noise, even with high-fidelity gates [175]. During a quantum calculation, these mistakes may spread and magnify until erroneous conclusions follow. Developed in 1995, quantum error correction addresses the issue resulting from the fragility of coherent quantum systems by use of active approaches [176]. Quantum error correction methods are thus important for encoding quantum information in a manner that is error-resistant and helps to overcome this constraint. Encoding each qubit of information over many physical qubits produces redundancy that enables the identification and repair of faults in quantum error correction [177]. Quantum error correction has significant overhead that calls for several physical qubits to guard

each logical qubit. Still another difficult obstacle confronting quantum computing is scalability. Solving difficult problems beyond the grasp of conventional computers requires the construction of quantum computers, including many qubits. As the system's complexity increases exponentially with the number of qubits, it is difficult to increase the number of qubits in a quantum computer while nevertheless retaining good fidelity and coherence. Furthermore, it is more difficult to monitor and regulate the interactions among qubits as the number of qubits rises, which increases error rates and lowers performance. New designs and manufacturing methods that can allow a lot of qubits while maintaining their quantum characteristics are needed in quantum computing. Whether it is challenging or absurdly difficult to govern big-scale quantum systems is still unknown [21, 178]. Increasing the qubits from hundreds to thousands presents a difficulty as existing systems cannot readily be expanded in size by sheer force [173]. Quantum computers and communications technologies will probably grow together as they share certain fundamental physics ideas, laboratory techniques, and similar hardware [179]. Though yet in its infancy, quantum computing has great potential to transform artificial intelligence [180, 181] medicine, materials science, and other domains. Long-standing proof-of-concept studies have shown the basic ideas of quantum computing—superposition, entanglement, and randomness—long ago [60]. Large-scale quantum computers will need development using quantum versions of tools, architectures, languages, compilers, and levels of abstraction [57]. Looking ahead, the advent of fault-tolerant quantum computing promises key advancements across industries, including drug discovery, financial modeling, optimization, and artificial intelligence. The potential to solve problems intractable for classical systems—such as simulating complex quantum systems, optimizing large-scale networks, and decrypting cryptographic codes—underscores the revolutionary impact of quantum computing. However, achieving these goals requires addressing current limitations while exploring novel applications that leverage the unique capabilities of quantum systems.

Hence, this analysis highlights the interconnected challenges and opportunities that define the current landscape of quantum computing. By synthesizing common themes and proposing interdisciplinary solutions, we aim to provide a structured framework for addressing the obstacles that hinder progress. As the field continues to evolve, collaborative innovation and resource-efficient strategies will be critical to unlocking the full potential of quantum computing and driving its adoption across scientific and industrial domains. This forward-looking perspective underscores the revolutionary potential of quantum computing while emphasizing the need for sustained efforts to overcome existing barriers.

5 | Conclusion

Quantum computing stands at the forefront of technological innovation, offering unprecedented computational capabilities rooted in the principles of quantum mechanics. This review has elucidated the foundational advancements in quantum hardware, algorithms, and applications, highlighting milestones such as quantum supremacy and the development of fault-tolerant

systems. Quantum algorithms, including Shor's, Grover's, and the HHL algorithm, have demonstrated profound potential in cryptography, optimization, and machine learning, while applications in drug discovery, precision medicine, financial modeling, and materials science underscore the transformative impact of quantum computing across industries. Despite these advancements, challenges such as qubit stability, noise mitigation, and scalability remain critical barriers to widespread adoption. The integration of quantum and classical systems, along with advancements in error correction techniques, will be pivotal in realizing the full potential of quantum computing. As interdisciplinary research continues to bridge theoretical and practical gaps, quantum computing is poised to address complex global challenges with unparalleled efficiency and precision. The future of this field lies in collaborative innovation, robust hardware development, and the creation of scalable quantum algorithms, ensuring its role as a cornerstone of next-generation computational technology.

Author Contributions

Christopher Columbus Chinnappan: conceptualization, writing – review and editing, validation. **Palani Thanaraj Krishnan:** software, writing – original draft, methodology, visualization. **Elakiya Elamaran:** data curation, resources. **Rajakumar Arul:** writing – review and editing. **T. Sunil Kumar:** formal analysis, supervision.

Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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